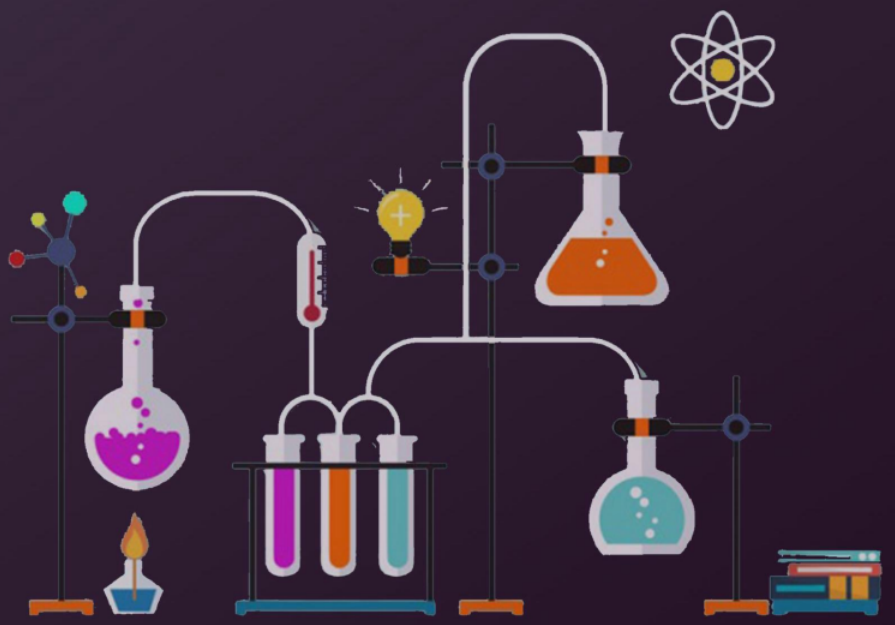
CHEMISTRY

Coordination Compound



SHEET

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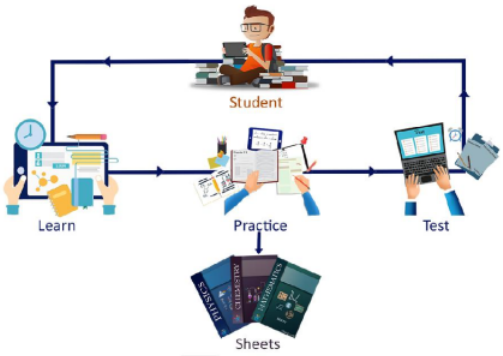
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Coordination Compounds	1,2	30		1,2
2	Ligands and Effect of Chelation	3-17	1-3,21-24	1,2	32
3	Synergic Bonding	18-21		3,4	
4	Organometallic Compounds and Sidgwick Rule	22-24	4,25	5,6	3,12-14
5	Werner's Theory	25,26		7-10	
6	IUPAC Nomenclature	27-34	5-7	11-14	
7	Structural Isomerism	35-38	26,27	15-19	4,29,30
8	Geometrical and Optical Isomerism in Square Planar Geometry	38-45	18,31,32	20-27	5,6,33
9	Stereoisomerism for M(a6) Type Complex	46,47	28,29	28-30	
10	Stereoisomerism for M(AA),M(AB) Type Complex	48			
11	VBT in Coordination Compounds	49-57	8-10,16, 17,33	31-37	15-19,23-25
12	CFT in octahedral geometry - Part 1				7
13	CFT in octahedral geometry - Part 2	58,59	19	38,39	
14	CFT in Tetrahedral Complexes		34		
15	CFT in Square Planar Geometry				
16	Question on CFT				
17	Stability of Coordination Compounds	60,61	11		20-22
18	Colour in Coordination Compounds	62			31,34
19	ACC	63	12-15,20	40-46	8-11,26-28, 35

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Coordination Chemistry

Q1. Which of the following is not a double salt but is a complex salt –
(A) $KCl.MgCl_2.6H_2O$
(B) $FeSO_4.(NH_4)_2SO_4.6H_2O$
(C) $K_2SO_4.Al_2(SO_4)_3.24H_2O$
(D) $4KCN.Fe(CN)_2$

Q2. In the complex $[Co(NH_3)_6]^{3+}$ the species acting as Lewis acid and Lewis bases are respectively.
(A) Co^{2+}, NH_3 (B) NH_3, Co
(C) Co^{3+}, NH_3 (D) NH_3, Co^{3+}

Ligands and Effect of Chelation

Q3. Triphenyl phosphine is -
(A) Neutral and monodentate ligand
(B) Neutral and tridentate ligand
(C) Uninegative and unidentate ligand
(D) Trinegative and tridentate ligand

Q4. In the complex ion $[Fe(EDTA)]^-$ the coordination number and oxidation state of central metal ion is :-
(A) C. N. = 6 O. N. = +3
(B) C.N. =1 O. N. = –1
(C) C. N. = 4 O. N. = +2
(D) C. N. = 3 O. N. = +3

Q5. The coordination number and oxidation number of the central metal ion in the complex $[Pt(en)_2]^{+2}$ is:-
(A) C. N. = 2,O. N. = +2
(B) C. N. = 6,O. N. = +4
(C) C. N. = 4,O. N. = +4
(D) C. N. = 4,O. N. = +2

Q6. Which of the following pair of ligand have same charge ?
(A) Sulphato & thio cyanato
(B) Oxalato & sulphito
(C) Nitrito & carbonato
(D) Dimethyl glyoximato & oxalato

Q7. Bidentate ligand is
(A) CN^-
(B) Ethylene diamine
(C) EDTA
(D) SCN^-

Q8. Species which is ligand?
(A) $NH_4^{\oplus}$ (B) $H_3O^{\oplus}$
(C) $N_2H_5^{\oplus}$ (D) H_2

Q9. The total number of Ligands attached to the central metal ion through coordinate bond is called -
(A) Valency of the metal ion
(B) Oxidation state of the metal ion
(C) Coordination number of metal ion
(D) None of the above

Q10. Co-ordination number and oxidation number of platinum in complex $[Pt(O_2)(en)_2Br]^{+1}$ are :-
(A) 4 and 3 respectively
(B) 4 and 2 respectively
(C) 6 and 4 respectively
(D) 6 and 2 respectively

Q11. The coordination number and oxidation number of the central metal ion in the complex $[Pt(en)_2]^{+2}$ is:-
(A) C. N. = 2,O. N. = +2
(B) C. N. = 6,O. N. = +4
(C) C. N. = 4,O. N. = +4
(D) C. N. = 4,O. N. = +2

Q12. Oxidation number of Mn in $(MnO_4)^-$ is
(A) –7 (B) +7
(C) +2 (D) –2

Q13. The oxidation state of Fe in $[Fe(CO)_5]$ is
(A) Zero (B) 1
(C) 2 (D) 2.5

Q14. Type of ligands co-ordination number and oxidation number of platinum in complex $[Pt(O_2)(en)_2Br]^{+1}$ are :-
(A) 2, 4 and 3 respectively
(B) 4, 4 and 2 respectively
(C) 3, 6 and 4 respectively
(D) 5, 6 and 2 respectively

Q15. In which of the following compounds, the transition metal is in oxidation state of zero ?
(A) $[Fe(H_2O)_3(OH)_3]$
(B) $[Ni(CO)_4]$
(C) $[Fe(H_2O)_6]SO_4$
(D) $[Co(NH_3)]_6Cl_3$

Q16. All ligands are –
(A) Lewis acid (B) Lewis base
(C) Neutral (D) None of these

Q17. The neutral ligand is –
(A) Chloro (B) Hydroxo
(C) Ammine (D) Oxalato

Synergic Bonding

Q18. Among the following metal carbonyls, the C–O bond order is lowest in :-
(A) $[Mn(CO)_6]^+$ (B) $[Fe(CO)_5]$
(C) $[Cr(CO)_6]$ (D) $|V(CO)_6|^-$

Q19. Which of the following species show(s) synergic bonding :
(A) $[Mo(CO)_6]$
(B) $[Mn(CO)_6]^-$
(C) $[Ni(CN)_4]^{-4}$
(D) All above species have synergic bonding

Q20. Compound which is represent minimum bond length of CO ?
(A) $[Fe(CO)_5]$ (B) $[Fe(CO)_5]^{\ominus}$
(C) $[Fe(CO)_5]^{\oplus}$ (D) $[Mn(CO)_5]$

Q21. In metal carbonyls, there is -
(A) No π bond between CO and metal atom
(B) Only σ bond between metal atom and CO molecules
(C) One σ and one π bond (back-donation) between metal atom and CO molecules
(D) The metal-carbon bonds does not exist at all

Orgonometallic Compounds and Sidgwik Rule

Q22. The EAN of iron in $K_4[Fe(CN)_6]$ is :
(A) 35 (B) 34
(C) 36 (D) 38

Q23. The EAN of cobalt in the complex ion $[Co(en)_2Cl_2]^+$ is :-
(A) 27 (B) 36
(C) 33 (D) 35

Q24. CH_3–Mg–Br is an organometallic compound due to :
(A) π – bond between C and Mg
(B) Mg – Br covalent bond
(C) σ – bond between C and Br
(D) σ – bond between C and Mg

Werner's Theory

Q25. Give the correct order of electrical conductivity of aqueous solution of following complexes :-
(A) $[Pt(NH_3)_6]Cl_4$
(B) $[Cr(NH_3)_6]Cl_3$
(C) $[Co(NH_3)_4Cl_2]$ Cl
(D) $K_2[PtCl_6]$
(A) A < B < C < D (B) A > B > D > C
(C) A < B < D < C (D) A > B < C < D

Q26. Which of the following has the highest molar conductivity solution?
(A) $[Pt(NH_3)_6]Cl_4$ (B) $[Pt(NH_3)_5Cl]Cl_3$
(C) $[Pt(NH_3)_4Cl_2]Cl_2$ (D) $[Pt(NH_3)_3Cl_3]Cl$

IUPAC Nomenclature

Q27. The I.U.P.A.C. name of $K_3[Ir(C_2O_4)_3]$ is :-
(A) Potassium tri oxalato Iridium (III)
(B) Potassium tri oxalato Iridate (III)
(C) Potassium tris (oxalato) Iridium (III)
(D) Potassium tris (oxalato) Iridate (II)

Q28. The correct name of the compound $[Cu(NH_3)_4](NO_3)_2$, according to IUPAC system is :-
(A) Cuprammonium nitrate
(B) Tetraamine copper (II) dinitrate
(C) Tetraammine copper (II) nitrate
(D) Tetraammine copper (I) dinitrate

Q29. The IUPAC name of the complex $[CrCl_2(H_2O)_4]NO_3$ is :-
(A) Dichlorotetraaquachromium (III) nitrate
(B) Tetra aqua dichloro chromium (III) nitrate
(C) Chromium tetra aqua dichloro nitrate
(D) Dichlorotetraaquachromium nitrate

Q30. The IUPAC name of $[Co(NH_3)_3ClBrNO_2]$ will be :-
(A) Triamminebromochloronitrocobaltate (III)
(B) Triamminebromochloronitrocobalt (III)
(C) Triamminebromonitrochlorocabalt (III)
(D) Triamminenitrochlorocobalt (III)

Q31. Which of the following complex is anion :-
(A) Fluoro pentaammine cobalt (III)
(B) Trioxalato ferrate (III)
(C) Penta Carbonyl iron (0)
(D) Dichloro diammine platinum

Q32. The IUPAC name of $K_2[Cr(CN)_2O_2(O)_2(NH_3)]$ is -
(A) Potassiumamminedicyanodioxoperoxo chromate (VI)
(B) Potassiumamminecyanoperoxodioxo chromium (VI)
(C) Potassiumamminecyanoperoxodioxo chromium (VI)
(D) Potassiumamminecyanoperoxodioxo chromate (IV)

Q33. Which of the following is correct IUPAC name of any complex compound.
(A) Tris(acetyl acetonato)iron(III) chloride
(B) Hexachloroplatinum(IV) tetraamminedicyanoplatinate(IV)
(C) Amminebromochloromethylamineplatinum(II)
(D) Cis dichloro(ethylenediamine)platinum(III)

Q34. The complex chloro diaquatriammine cobalt (III) chloride is represented as :-
(A) $[Co(NH_3)_3(H_2O)_3]Cl_2$
(B) $[Co(NH_2)_3(H_2O)_2]Cl_2$
(C) $[CoCl(NH_3)_3(H_2O)_2]Cl_3$
(D) $[CoCl(NH_3)_3(H_2O)_2]Cl_2$

Structural Isomerism

Q35. Which one of the following is an example of coordination isomerism?
(A) $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$
(B) $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5ONO]Cl_2$
(C) $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$
(D) $[Cr(NH_3)_6][Co(CN)_6]$ and $[Co(NH_3)_6][Cr(CN)_6]$

Q36. In coordination compounds, the hydrate isomers differ –
(A) In the number of water molecules of hydration only
(B) In the number of water molecules only present as ligands
(C) Both (A) and (B)
(D) In their coordination number of the metal atom

Q37. Which of the following pair of complex is co-ordination isomers :
(A) $[Co(NH_3)_5Br]SO_4$ & $[Co(NH_3)_5SO_4]Br$
(B) $[Pt(NH_3)_4][PtCl_4]$ & $[Pt(NH_3)_2Cl_2][Pt(NH_3)_2Cl_2]$
(C) $[Co(H_2O)_6]Cl_3$ & $[Co(H_2O)_5Cl]Cl.2H_2O$
(D) None of these

Q38. $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5(ONO)]Cl_2$ are related to each other as :

(A) Geometrical isomers

(B) Optical isomers

(C) Linkage isomers

(D) Co-ordination isomers

Geometrical & Optical Isomerism in Square Planar Geometry

Q39. A complex compound which can show the optical isomerism but not Geometrical isomerism is :-

(A) $[Co(en)_2Cl_2]^+$ (B) $[Co(NH_3)_5Cl]^{+2}$

(C) $[Co(OX)_3]^{-3}$ (D) $[Pt(NH_3)_2Cl_2]$

Q40. Which of the following set of isomerism is wrong :

(A) Cis-$[Co(gly)_2Cl_2]^-$ – optical isomerism

(B) $[Zn(NH_3)_3Cl]^+$ – Geometrical isomerism

(C) $[Fe(H_2O)_6]Cl_3$ – Hydrate isomerism

(D) Co-ordination isomers

Q41. Which types of Isomerism can be exhibited by $[Co(NH_3)_2(en)(NO_2)_2]^+$

(A) Geometrical only

(B) Both geometrical and linkage Isomerism

(C) Linkage Isomerism only

(D) Both Geometrical and Ionisation Isomerism.

Q42. Which one of the following can show GI & OI both ?

(A) $[Co(NH_3)_3Cl_3]$ (B) $[Co(NH_3)_4Cl_2]^+$

(C) $[Co(en)_3]^{+3}$ (D) $[Co(en)_2Cl_2]^+$

Q43. Which of the following complex or complex ion can have optically active isomer -

(A) $[Co(OX)_3]^{3-}$

(B) $[Co(H_2O)_4Cl_2]$ $Cl.2H_2O$

(C) $Na_2[ZnCl_4]$

(D) $[Pt(en)_2]^{2+}$

Q44. In $[Cr(C_2O_4)_3]^{2-}$, the isomerism shown is :-

(A) ligand (B) optical

(C) geometrical (D) ionisation

Q45. Geometrical isomerism in coordination compounds is exhibited by -

(A) Square planar and tetrahedral complexes

(B) Square planar and octahedral complexes

(C) Tetrahedral and octahedral complexes

(D) Square planar, tetrahedral and octahedral complexs

Sterioisomerism for M(a6) Type Complex

Q46. Which of the following statement is correct regarding the $[Ma_2b_2c_2]$ type of complex (where a, b, c are monodentate ligand having no chiral centre)

(A) cis form is optically active

(B) trans forms are optically inactive

(C) Total number of geometrical isomers is 5

(D) All of these

Q47. Which of the following complexes can exist as enantiomers?

(A) $[Cr(NH_3)_2Cl_4]^-$ (B) $[Co(NH_3)_5Br]^{2+}$

(C) $[FeCl_2(NCS)_2]^{2-}$ (D) $[Co(en)_3]^{3+}$

Sterioisomerism for M(AA), M(A,B) Type Complex

Q48. Which of the following complex compound is optically inactive :

(A) $[M(AA)_2cd]$

(B) [M(AB)(CD)(EF)]

(C) $[M(AA)_2]$

(D) Mabcd(tetrahedral)

VBT in Coordination Compounds

Q49. d^2sp^3 hybridisation leads to :-

(A) Hexagonal shape

(B) Trigonal bipyrimidal

(C) Octahedral shape

(D) Tetrahedral shape

Q50. What type of hybridization is involved in $[Fe(CN)_6]^{3-}$?

(A) d^2sp^3 (B) dsp^2

(C) sp^3d^2 (D) dsp^3

Q51. No. of unpaired e^- present in $[Mn(NH_3)_6]N_2$:-

(A) 5 (B) 7

(C) 1 (D) 3

Q52. In which of following pairs of species the number of unpaired electrons are same :-

(A) $[CoF_6]^{3-}$, $[FeF_6]^{3-}$

(B) $[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$

(C) $[Fe(CN)_6]^{3-}$, $[Ni(CN)_4]^{2-}$

(D) $[CoF_6]^{3-}$, $[Fe(H_2O)_6]^{2+}$

Q53. Which of the following is outer orbital complex?

(A) $[FeF_6]^{-3}$ (B) $[Fe(CN)_6]^{-3}$

(C) $[Cr(NH_3)_6]^{+3}$ (D) $[Mn(CN)_6]^{-3}$

Q54. A Cuprammonium ion $[Cu(NH_3)_4]^{2+}$ is :-

(A) Tetrahedral

(B) Square planar

(C) Triangular

(D) Octahedral

Q55. The complex which has no 'd' electron in the central metal atom is :-

(A) MnO_4^- (B) $[Co(NH_3)_6]^{3+}$

(C) $[Fe(CN)_6]^{3-}$ (D) $[Cr(H_2O)_6]^{3+}$

Q56. The species having tetrahedral shape is :-

(A) $[PdCl_4]^{2-}$ (B) $[Ni(CN)_4]^{2-}$

(C) $[Pd(CN)_4]^{2}$ (D) $[NiCl_4]^{2-}$

Q57. Which of the following pair differ in geometry and resemble in magnetic behaviour.

(A) $[Ni(CO)_4]$ & $[Ni(CN)_4]^{2-}$

(B) $[Be(H_2O)_4]^{2+}$ & $[Zn(CN)_4]^{2-}$

(C) $[Fe(CN)_6]^{3-}$ & $[FeF_6]^{3-}$

(D) $[Fe(CO)_5]$ & PCl_5

CFT in Octahedral Geometry

Q58. Which of the following hybrid state is associated with low spin octahedral complex ?

(A) dsp^2 (B) sp^3d^2

(C) d^2sp^3 (D) sp^3d

Q59. In an octahedral crystal field, the t_2g orbitals are-

(A) Raised in energy by $0.4\ \Delta_0$

(B) Lowered in energy by $0.4\ \Delta_0$

(C) Raised in energy by $0.6\ \Delta_0$

(D) Lowered in energy by $0.6\ \Delta_0$

Stability of Coordination Compound

Q60. Select the correct answer :-

(A) $[Pt(en)_3]SO_4$ shows optical and geometrical isomerism

(B) $K_3[FeF_6]$ is less stable than $K_3[Fe(CN)_6]$

(C) In $[Pt(en)_2Cl_2]$ Coordination number is = 4

(D) All of them

Q61. Select the correct order of stability :-

(A) $[Cr(H_2O)_6]^{+3} < [Cr(NH_3)_6]^{+3} < [Cr(CN)_6]^{-3}$

(B) $[Cr(NH_3)_4(en)]^{+3} < [Cr(NH_3)_2(en)_2]^{+3} < [Cr(en)_3]^{+3}$

(C) $[Co(H_2O)_6]^{+2} < [Co(H_2O)_6]^{+3}$

(D) All of them

Colour in Coordination Compound

Q62. Which of the following colour frequency is absorbed by aquated Ni^{2+} ion-

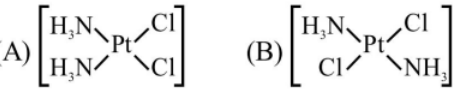
(A) red (B) blue

(C) green (D) none of these

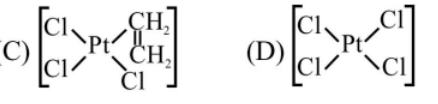
ACC

Q63. Which of the following is considered as an anticancer species :-

(A)

(B) 

(C)

(D) 

Exercise - 1A
Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. SCN ligand if N is attached to central atom, the Incorrect name of ligand is -
(A) Thiocyanato-N (B) Cyanato-N
(C) Thiocyanato-S (D) Cyanato-S

Q2. Which order is incorrect in spectrochemical series of ligands -
(A) $Cl^- < F^- < C_2O_4^{2-} < NO_2^- < CN^-$
(B) $CN^- < C_2O_4^{2-} < Cl^- > NO_2^- < F^-$
(C) $C_2O_4^{2-} < F^- < Cl^- > NO_2^- < CN^-$
(D) $F^- < Cl^- < NO_2^- < CN^- < C_2O_4^{2-}$

Q3. Which of the following ligands is/are chelating agent –
(A) EDTA (B) en
(C) Oxalate (D) Pyridine

Q4. Which of the following contain metal carbon bond -
(A) $Al(OC_2H_5)_3$ (B) C_2H_5MgBr
(C) $K[Pt(C_2H_4)Cl_3]$ (D) $Ni(CO)_4$

Q5. The incorrect IUPAC name for $K_2[OsCl_5N]$ is/are -
(A) Potassium pentachloroazidoosmate (VIII)
(B) Potassium pentachloroazidoosmate (VI)
(C) Potassium pentachloronitridoosmate (VI)
(D) Potassium nitroosmate (III)

Q6. The complex compound $[Co(NH_3)_3NO_2ClCN]$ is not named as -
(A) TriammineChlorocyanonitro cobalt (III)
(B) Nitrochlorocyanotriammine cobalt (III)
(C) Cyanonitrochlorotriammine cobalt (III)
(D) Triamminenitrochlorocyano cobalt (III)

Q7. Which of the following complex is/are cation :-
(A) Fluoropentaamminecobalt (III)
(B) Trioxalatoferrate (III)
(C) Pentacarbonyliron (0)
(D) Dichlorodiammineplatinum

Q8. The complex ion which has 'd' electrons in the central metal atom is –
(A) $[MnO_4]^-$
(B) $[Co(NH_3)_6]^{3+}$
(C) $[Fe(CN)_6]^{3-}$
(D) $[Cr(H_2O)_6]^{3+}$

Q9. Atomic numbers of Cr and Fe are 24 and 26 respectively, which of the following is diamagnetic?
(A) $Cr(CO)_6$
(B) $Fe(CO)_5$
(C) $Fe(CN)_6^{4-}$
(D) $Cr(NH_3)_6^{3+}$

Q10. A magnetic moment of 1.73 BM will be shown by :
(A) $[Zn(CN)_4]^{2-}$
(B) $TiCl_3$
(C) $[Fe(CN)_6]^{4-}$
(D) $[Cu(NH_3)_4]^{2+}$

Q11. $[Cu(CN)_4]^{2-}$ is more stable that $[Cu(NH_3)_4]^{2+}$ because :-
(A) stability constant of $[Cu(CN)_4]^{2-}$ is greater than $[Cu(NH_3)_4]^{2+}$
(B) CN^- is stronger ligand than NH_3
(C) NH_3 is stronger than CN^-
(D) none

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 12 to 15

Coordination compounds life play a vital role in our lives. The importance of these compounds can be realised from the fact that life would not have been possible without the existence of chlorophyll in plants and heamoglobin in the blood of the animals. The field of such compounds has expanded very fast in recent years and coordination compounds are playing important role in analytical chemistry, polymerisation reactions, metallurgy and refining of metals, organic synthesis, electroplating, biochemistry water purification, textile dyeing and bacteriology.

Q12. The hardness of water is estimated by -
(A) Conductivity method
(B) EDTA method
(C) Titrimetric method
(D) Distillation method

Q13. A reagent used for identifying nickel ion is -
(A) Potassium ferrocyanide
(B) Phenolphthalein
(C) Dimethyl glyoxime
(D) EDTA

Q14. Lead poisoning in the body can be removed by -
(A) Zeise's salt
(B) Ferrocene
(C) Cis-platin
(D) $[Ca(EDTA)]^{2-}$

Q15. Which is used in cancer chemotherapy -
(A) Cis-platin
(B) Zeise's salt
(C) Ferrocene
(D) Fischer's salt

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q16. Match List-I with List-II and select the correct answer using codes given below the lists –

List-I Metal ions	**List-II** Magnetic moment(B.M.)
(A) Cr^{3+}	i. $\sqrt{35}$
(B) Fe^{2+}	ii. $\sqrt{30}$
(C) Ni^{2+}	iii. $\sqrt{24}$
(D) Mn^{2+}	iv. $\sqrt{15}$
	v. $\sqrt{8}$

Code :

	(1)	(2)	(3)	(4)
(A)	i	iii	v	iv
(B)	ii	iii	vi	i
(C)	iv	iii	v	i
(D)	iv	v	iii	i

Q17. Match List–I (Complex ions) with List–II (Number of Unpaired Electrons) and select the correct answer using the codes given below the lists :-

List –I (Complex ions)	**List II (Number of Unpaired Electrons)**
A. $[CrF_6]^{4-}$	i. One
B. $[MnF_6]^{4-}$	ii. Two
C. $[Cr(CN)_6]^{4-}$	iii. Three
D. $[Mn(CN)_6]^{4-}$	iv. Four
	v. Five

Code :

	A	B	C	D
(A)	iv	i	ii	v
(B)	ii	v	iii	i
(C)	iv	v	ii	i
(D)	ii	i	iii	v

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q18.

Column-I	**Column-II**
(A) $[CoCl_3(NH_3)_3]$	(i) show facial form
(B) $[Cr(Ox)_3]^{3-}$	(ii) optically active structure
(C) $[CrCl_2(Ox)_2]$	(iii) Trans form is optically inactive
(D) $RhCl_3(Py)_3]$	(iv) show meridian form

Q19.

Column-I	**Column-II**
(A) d^2sp^3	(i) outer orbital octahedral
(B) sp^3d^2	(ii) High spin complex
(C) $K_3[Fe(CN)_6]$	(iii) Low spin complex
(D) $[Fe(H_2O)_6]Cl_3$	(iv) octahedral

Q20. Match the column I with column II and mark the correct option from the codes given below :-

	Column I (Complex species)		**Column II** (Isomerism)
i.	$[Co(NH_3)_4Cl_2]^+$	P.	Optical
ii.	cis-$[Co(en)_2Cl_2]^+$	q.	Ionisation
iii.	$[Co(NH_3)_5(NO_2)]Cl_2$	r.	Coordination
iv.	$[Co(NH_3)_6][Cr(CN)_6]$	s.	Geometrical
		t.	Linkage

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q21. Sum of coordination number and oxidation number of Cr in $K_3[Cr(C_2O_4)_3]$ are, respectively -

Q22. The co-ordination number of Fe in the complex ion $K_3[Fe(CN)_5NO]$ is -

Q23. What is the value of x in the carbonyl $[Fe(CO)_x]$?

Q24. How many isomerism are possible for the complex, $[Cr(NH_3)_4ClBr]Br$?

Q25. Find out some of oxidation no. of Cr and Co in the following compounds. $[Cr(NH_3)_6]$ $[Co(C_2O_4)_3]$

Q26. Theoritically the No. of geometrical isomers expected for octahedral complex [Mabcdef] is -

Q27. What is the crystal field stabilization energy, in terms of Δ, for Zn^{2+} ion in a tetrahedral complex ?

Q28. How many isomers are possible for the complex ion, $[Cr(NH_3)(OH)_2Cl_3]^{2-}$?

Q29. The number of chloride ions produced in solution of $CoCl_3.5NH_3$ are

ASSERTION AND REASON

These questions contains, Statement-I (assertion) and Statement-II (reason).

(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I.

(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I.

(C) Statement-I is True, Statement-II is False.

(D) Statement-I is False, Statement-II is True.

Q30. **Statement-I :** $[Fe(CO)_5]$ is dimagnetic complex.
Statement-II : In the given complex oxidation state of Iron is zero.

Q31. **Statement-I :** $[Co(NO_2)_3(NH_3)_3]$ does not show optical isomerism.
Statement-II : It has plane of symmetry.

Q32. **Statement-I :** $[Ni(CN)_4]^{-2}$ has zero unpaired electron while that of $[NiCl_4]^{-2}$ has two unpaired e^-.
Statement-II : $[Ni(CN)_4]^{-2}$ has strong crystal field while $[NiCl_4]^{-2}$ has weak crystal field.

Q33. **Statement-I :** Trans $[CoCl_2(en)_2]^+$ is optically inactive.
Statement-II : It has plane of symmetry.

Q34. **Statement-I :** $K_2SO_4.Al_2(SO_4)_3.24H_2O$ is a double salt compound.
Statement-II : It ionises to give a complex ion.

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. To form a coodination bond, one needs a ligand. Which of the following species cannot be a ligand -

(i) NH_4^+ (ii) NO^+

(iii) $C_5H_5\overset{\square\square}{N}$

(A) i only (B) i & ii only

(C) i & iii only (D) i, ii & iii only

Q2. The donor atoms in EDTA are -

(A) Two N and Two O

(B) Two N and four O

(C) Four N and Two O

(D) Three N and three O

Q3. Among the following paramagnetic carbonyl species is -

(A) Mn $H(CO)_5$

(B) $[Mn(CO)_5]^-$

(C) $Mn_2(CO)_{10}$

(D) $V(CO)_6$

Q4. Select the true statement from the following for metal carbonyls ?

(A) π-backbonding strengthens M–C bond order as well as CO bond order

(B) π-backbonding weakens M–C bond order as well as CO bond order

(C) π-backbonding weakens M–C bond order but strongthens CO bond order

(D) π-backbonding strengthens M–C bond order & weakens CO bond order

Q5. The effective atomic number of cobalt in the complex $[Co(NH_3)_6]^{3+}$ is :-

(A) 36 (B) 33

(C) 24 (D) 30

Q6. Select the odd one w.r.t. EAN :-

(A) $[Fe(CO)_5]$ (B) $[Cr(CN)_6]^{-3}$

(C) $[Mn_2(CO)_{10}]$ (D) $[Fe(CN)_6]^{-4}$

Q7. In the compound $CoCl_3 \cdot 5NH_3$

(A) all the Cl show primary valency (PV) only

(B) two Cl show (PV) and one Cl secondary valency (SV)

(C) two Cl show (PV) and one Cl (PV) as well as (SV)

(D) all the Cl show (SV)

Q8. One mole of the complex compound $Co(NH_3)_5Cl_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of $AgNO_3$ solution to yield two moles of AgCl(s). The structure of the complex is :-

(A) $[Co(NH_3)_5Cl]Cl_2$

(B) $[Co(NH_3)_3Cl_3]2NH_3$

(C) $[Co(NH_3)_4Cl_2]Cl.NH_3$

(D) $[Co(NH_3)_4Cl]Cl_2.NH_3$

Q9. The wrong statement regarding complex $PtCl_4.3NH_3$ is :-

(A) It gives two ions per molecule in a solution

(B) Secondary valency of Pt^{+4} is three

(C) It gives 1 mole of AgCl with $AgNO_3$

(D) Three Cl^- satisfying primary as well as secondary valencies

Q10. A co-ordination complex of cobalt has molecular formula containing five ammonia molecules, one nitro group and two chlorine atoms for one cobalt atom. One mole of this compound produces three mole ions in an aqueous solution. In reacting this solution with excess of silver nitrate solution, two moles of AgCl get precipitated. The ionic formula of this complex would be -

(A) $[(Co(NH_3)_4.NO_2Cl].[(NH_3)Cl]$

(B) $[(Co(NH_3)_5Cl].[Cl(NO_2)]$

(C) $[(Co(NH_3)_5(NO_2)]Cl_2$

(D) $[(Co(NH_3)_5].[(NO_2)_2Cl_2]$

Q11. The IUPAC name for $[Co(NH_3)_6]\,[Cr(CN)_6]$ is :-

(A) Hexaammine cobalt (III) hexacyanochromate (III)

(B) Hexacyanochromium cobalt hexaammine (VI)

(C) Hexaammine cobalt (III) hexacyanochromium (VI)

(D) Hexacyanochromium (III) hexaammine cobalt (III)

Q12. Which of the following is incorrect IUPAC naming-

(A) $[Co(NH_3)_5NO_2]_3\,[Co(NO_2)_6]_2$ pentaamminenitrito-N-cobalt(III) hexanitrito-N-cobaltate(III)

(B) $H_2[OsCl_5N]$ hydrogenhexacloridoosmate(IV)

(C) $K_2[OsCl_5N]$ Potassium pentachloridonitridoosmate(VI)

(D) $[Cr(H_2O)_6]^{3+}$ hexaaquachromium(III)ion

Q13. The correct name of $[Pt(NH_3)_4Cl_2]\,[PtCl_4]$ is :-

(A) Tetraammine dichloro platinum (IV) tetrachloro platinate (II)

(B) Dichloro tetra ammine platinium (IV) tetrachloro platinate (II)

(C) Tetrachloro platinum (II) tetraammine platinate(IV)

(D) Tetrachloro platinum (II) dichloro tetraamine platinate (IV)

Q14. The incorrect IUPAC name of compound is-

(A) $[BF_4]^-$ Tetrafluoridoborate(III) ion

(B) $H_2[SiF_6]$ Hexafluoridosilicic(IV) acid

(C) $[Pt(NH_3)_2(NO_2)_2]$ Dinitro-Ndiammine platinum(II)

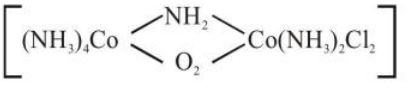
(D) None of these

Q15. Which of the following complex can shows coordination isomerism :-

(A) $[Pt(H_2O)_6]\,(SO_4)_2$

(B) $[Co(en)_3]\,[Cr(CN)_6]$

(C) $[Co(NH_3)_5\,ONO]Cl_2$

(D) 

Q16. The type of isomerism shown by $[Co(en)_2(NCS)_2]Cl$ and $[Co(en)_2(NCS)Cl]$ NCS is :

(A) Co-ordination (B) Ionisation

(C) Linkage (D) All of these

Q17. Which of the following pair of complex is co-ordination isomers :

(A) $[Co(NH_3)_5Br]SO_4$ & $[Co(NH_3)_5SO_4]Br$

(B) $[Pt(NH_3)_4]\,[PtCl_4]$ & $[Pt(NH_3)_2Cl_2]\,[Pt(NH_3)_2Cl_2]$

(C) $[Co(H_2O)_6]Cl_3$ & $[Co(H_2O)_5Cl]Cl.2H_2O$

(D) None of these

Q18. Two complex compounds has same molecular formula but different structural formula shows isomerism. If number of water molecules are different inside coordination sphere shows hydrated isomerism and gives different type of ions in aquous solution then shows ionisation isomerism ligands are exchanged in between cationic and anionic complex shows coordination isomerism. When ligand attached by different donar atoms then shows linkage isomerism. When ligands inter change in bridge compound then coordination isomerism and stoichiometric ratio of ions or molecules are same, then shows polymer isomerism.

The pair of complex compounds $[Cr(H_2O)_4Cl_2].Cl2H_2O$ and $[Cr(H_2O)_5Cl]Cl_2.H_2O$ are an example of :-

(A) Linkage isomerism

(B) Ionisation isomerism

(C) Coordination isomerism

(D) Hydrate isomersim

Q19. The two compounds pentaammine sulphatocobalt(III) bromide and pentaammine sulphatocobalt(III) chloride represent.

(A) Linkage isomerism

(B) Ionization isomerism

(C) Coordination isomerism

(D) No isomerism

Q20. Isomerisms exhibited by $[Cr(NH_3)_2(H_2O)_2Cl_2]^+$ are :

(A) ionisation, optical

(B) hydrate, optical

(C) geometrical, optical

(D) coordinate, geometrical

Q21. Out of the following which will not show geometrical isomerism :-

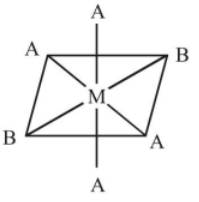
(A) $[Pt(NH_3)_2(H_2O)_2]^{+2}$

(B) $[Cr(NH_3)_5Cl]Cl_2$

(C) $[Co(en)_2Cl_2]Cl$

(D) $[Co(NH_3)_4Cl_2]Cl$

Q22. The isomer :-

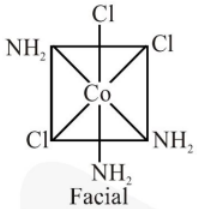


can be marked as -

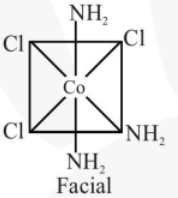
(A) Cis isomer (B) Leavo isomer

(C) Dextro isomer (D) Trans isomer

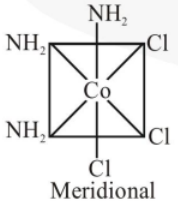
Q23. Select the correct representation ?

(A)


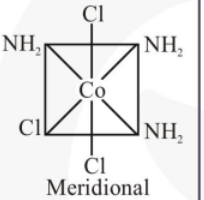
Facial

(B)


Facial

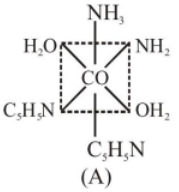
(C)


Meridional

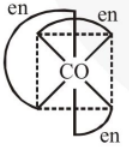
(D)


Meridional

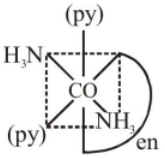
Q24. Which of the following will be optically inactive?



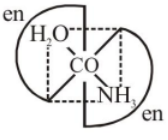
(A)



(B)



(C)



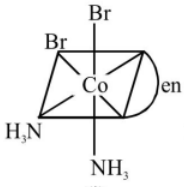
(D)

(A) A only (B) A & D

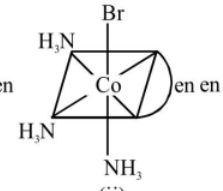
(C) A, C & D (D) All of these

Q25. Three arrangements are shown for the complex, $[Co(en)(NH_3)_2Br_2]^+$.

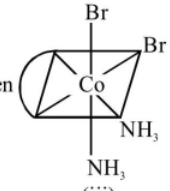
Which one is the wrong statement ?



(i)



(ii)



(iii)

(A) I and II are geometrical isomers

(B) II and III are optical isomers

(C) I and III are optical isomers

(D) II and III are geometrical isomers

Q26. Which of the following is opticaly action :

(a) $[Pt\ FCl\ Br\ I]^{2-}$ (b) $[Co(en)_3]^{3+}$

(c) $[Co(gly)_3]^{3+}$ (d) $[Zn\ FCl\ Br\ I]^{2-}$

(A) a,b, c, d (B) b,c,d

(C) c,d (D) b,c

Q27. Green Coloured $[Ni(PPh_2\ et)_2\ Br_2]$ has magnetic moment of 2.82 BM the geomatry and No. of isomers are possible for complex.

(A) Square planar and one

(B) Tetrahedral and two

(C) Square Planar and two

(D) Tetrahedral and zero

Q28. Which of the following statement is correct regarding the $[Ma_2b_2c_2]$ type of complex (where a, b, c are monodentate ligand having no chiral centre).

(A) cis form is optically active

(B) trans forms are optically inactive

(C) Total number of geometrical isomers is 5

(D) All of these

Q29. Which one of the following octahedral complex will not show geomatric isomerism (A and B are monodentate ligands) : -

(A) MA_5B (B) MA_2B_4

(C) MA_3B_3 (D) MA_2B_2

Q30. Which of the following planar complex is diamagnetic and show stereo isomerism :

(A) $[Ni(gly)_2]$ (B) $[Pt(NH_3)_3Cl]^+$

(C) $[Co(H_2O)_3Cl_3]$ (D) $[Co(en)_3]^{3+}$

Q31. Which of the following is expected to be a paramagnetic complex ?
(A) $[Ni(H_2O)_6]^{2+}$ (B) $[Ni(CO)_4]$
(C) $[Zn(NH_3)_4]^{2+}$ (D) $[Co(NH_3)]^{3+}$

Q32. Which one of the following complexes is an outer orbital complex :-
(A) $[Co(NH_3)_6]^{3+}$ (B) $[Mn(CN)_6]^{4-}$
(C) $[Fe(CN)_6]^{4-}$ (D) $[Ni(NH_3)_6]^{2+}$
(Atomic nos : Mn = 25, Fe = 26, Co = 27, Ni = 28)

Q33. Species which represent tetrahedral shape is ?
(A) $[PdCl_4]^{2-}$ (B) $[Ni(CN)_4]^{2-}$
(C) $[Pt(CN)_4]^{2-}$ (D) $[NiCl_4]^{2-}$

Q34. Among $Ni(CO)_4$, $[Ni(CN)_4]^{2-}$ and $NiCl_4^{2-}$ -
(A) $Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$ are diamagnetic and $NiCl_4^{2-}$ is paramagnetic
(B) $Ni(CO)_4$ and $NiCl_4^{2-}$ are diamagnetic and $[Ni(CN)_4]^{2-}$ is paramagnetic
(C) $Ni(CO)_4$ is diamagnetic and $[Ni(CN)_4]^{2-}$ and $NiCl_4^{2-}$ are paramagnetic
(D) $NiCl_4^{2-}$ and $[Ni(CN)_4]^{2-}$ are diamagnetic and $Ni(CO)_4$ is paramagnetic.

Q35. Among $[Ni(CO)_4]$, $[Ni(CN)_4]^{2-}$, $[NiCl_4]^{2-}$ species, the hybridization states at the Ni atom are, respectively :-
(A) sp^3, sp^3, dsp^2 (B) dsp^2, sp^3, sp^3
(C) sp^3, dsp^2, dsp^2 (D) sp^3, dsp^2, sp^3

Q36. Among the complex ions given below which is/are outer-orbital complex :

$[Co(CN)_6]^{3-}$	$[Fe(H_2O)_6]^{2+}$	$[FeF_6]^{3-}$	$[CoF_6]^{3-}$
I	II	III	IV

(A) II, III, IV (B) II, III only
(C) I , IV only (D) II only

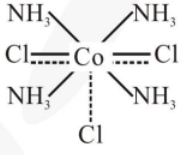
Q37. The pair of compounds having the same hybridisation for the central atom is :-
(A) $[Cu(NH_3)_4]^{2+}$ and $[Ni(NH_3)_4]^{2+}$
(B) $[NiCl_4]^{2-}$ and $[PtCl_4]^{2-}$
(C) $[Cu(NH_3)_4]^{2+}$ and $[Zn(NH_3)_4]^{2+}$
(D) $[Ni(NH_3)_6]^{3+}$ and $[Co(H_2O)_6]^{3+}$

Q38. For the $t_{2g}^6\ e_g^2$ system, the value of magnetic moment (μ) is :
(A) 2.83 B.M. (B) 1.73 B.M.
(C) 3.87 B.M. (D) 4.92 B.M.

Q39. Atomic number of Mn, Fe, Co are 25, 26 & 27 respectively. Which of the following inner orbital octahedral complex ion are diamagnetic :-
(A) $[Co(NH_3)_6]^{+3}$ (B) $[Mn(CN)_6]^{-3}$
(C) $[Fe(CN)_6]^{-4}$ (D) $[Fe(CN)_6]^{-3}$
(A) A, C (B) A, B, C
(C) A, C, D (D) A, D

Q40. The shape of water molecule is same as that of :
(A) C_2H_2 (B) CO_2
(C) NH_3 (D) Cl_2O

Q41. Select the correct statement about the given complex -



(A) It has four ions in aquous solution.
(B) Primary valency of cobalt is six.
(C) It give one mole of AgCl ppt on reaction with excess of $AgNO_3$.
(D) It has one primary valency

Q42. Which of the following will give maximum number of isomers ?
(A) $[Co(NH_3)_4Cl_2]$
(B) $[Ni(en)(NH_3)_4]^{2+}$
(C) $[Ni(C_2O_4)\ (en)_2]^{2-}$
(D) $[Cr(SCN)_2\ (NH_3)_4]^{+}$

Q43. Which of the following statement are not correct ?
(A) $[Ni(CO)_4]$ is tetrahedral and diamagnetic
(B) $[Co(NH_3)_3Cl_3]$ represent 2 geometrical isomers
(C) $[Fe(NH_3)_6]\ SO_4$ represent low spin complex and paramagnetic nature
(D) $[Co(Ox)_3]^{3-}$ represent optical isomerism

Q44. Which of the following is more paramagnetic :-
(A) $K_4[Fe(CN)_6]$ (B) $K_3[Fe(CN)_6]$
(C) $K_3[CoF_6]$ (D) $K_3[Co(CN)_6]$

Q45. For $K_3[CoF_6]$, incorrect statement is :
(A) It is high spin complex
(B) Its magnetic moment is $\sqrt{24}$ BM
(C) Primary valency of Co is six
(D) Hybridisation state of CoF_6^{3-} is sp^3d^2

Q46. The correct order for the wavelength of absorption in the visible region is –
(A) $[Ni(NO_2)_6]^{4-} < [Ni(NH_3)_6]^{2+} < [Ni(H_2O)_6]^{2+}$
(B) $[Ni(NO_2)_6]^{4-} < [Ni(H_2O)_6]^{2+} < [Ni(NH_3)_6]^{2+}$
(C) $[Ni(H_2O)_6]^{4-} < [Ni(NH_3)_6]^{2+} < [Ni(NO_2)_6]^{4-}$
(D) $[Ni(NH_3)_6]^{2+} < [Ni(H_2O)_6]^{2+} < [Ni(NO_2)_6]^{4-}$

Exercise - 2A *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The correct statement regarding complex $PtCl_4.3NH_3$ is :-
(A) It gives two ions per molecule in a solution
(B) Secondary valency of Pt^{+4} is three
(C) It gives 1 mole of AgCl with $AgNO_3$
(D) Three Cl^- satisfying primary as well as secondary valencies

Q2. Which of the following statements can be applicable to Ziese's salt ?
(A) Its aqueous solution gives test for Cl^- ion
(B) Its IUPAC name is potassium trichlorido (η^2 – ethene) platinate (II)
(C) Its formula is $K[Pt(C_2H_4)C\ell_3]$
(D) It contains one Pt^{4+} ion

Q3. Select the following which follow EAN rule :-
(A) $[Fe(CO)_5]$ (B) $[Cr(CN)_6]^{-3}$
(C) $[Mn_2(CO)_{10}]$ (D) $[Fe(CN)_6]^{-4}$

Q4. $[(C_6H_5)_2Pd(SCN)_2]$ and $[(C_6H_5)_2Pd(NCS)_2]$ are not -
(A) Linkage isomers
(B) Co-ordination isomers
(C) Ionisation isomers
(D) Geometrical isomers

Q5. Which one of the following can show GI but not OI?
(A) $[Co(NH_3)_3Cl_3]$ (B) $[Co(NH_3)_4Cl_2]^+$
(C) $[Co(en)_3]^{+3}$ (D) $[Co(en)_2Cl_2]^+$

Q6. The compound that exhibit GI is/are –
(A) $[Pt(en)Cl_2]$ (B) $[Pt(en)_2]Cl_2$
(C) $[Pt(en)_2Cl_2]Cl_2$ (D) $[Pt(NH_3)_2Cl_2]$

Q7. If $\Delta_0 < P$, the incorrect electronic configuration for d^4 system will be –
(A) $t_{2g}^4 e_g^0$ (B) $t_{2g}^3 e_g^1$
(C) $t_{2g}^0 t_g^4$ (D) $t_{2g}^2 e_g^2$

Q8. Which can't gives only 25% mole of AgCl, when reacts with $AgNO_3$:-
(A) $PtCl_2 . 4NH_3$ (B) $PtCl_4 . 5NH_3$
(C) $PtCl_4 . 4NH_3$ (D) $PtCl_4 . 3NH_3$

Q9. Which of the following planar complex is diamagnetic and show stereo isomerism :
(A) $[Ni(gly)_2]$ (B) $[Pt(NH_3)_2Cl_2]$
(C) $[Co(H_2O)_3Cl_3]$ (D) $[Co(en)_3]^{3+}$

Q10. For complex ions $[Fe(CN)_6]^{3-}$ and $[Fe(CN)_6]^{4-}$ the incorrect is :
(A) Are both octahedral and paramagnetic
(B) Are both octahedral and diamagnetic
(C) Have same structure but different magnetic character
(D) None of these

Q11. Which of the following statement are correct ?

(A) $[Ni(CO)_4]$ is tetrahedral and diamagnetic

(B) $[Co(NH_3)_3Cl_3]$ represent 2 geometrical isomers

(C) $[Fe(NH_3)_6]$ SO_4 represent low spin complex and paramagnetic nature

(D) $[Co(Ox)_3]^{3-}$ represent optical isomerism

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 12 to 14

On the basis of stability of complex entity in the solution, complexes may be of two types, perfect and imperfect complexes. The stability depends upon the extent of complex may also be defined as measure of resistance to the replacement of a ligand by some other lignad. The dissociation of a complex may be expressed as $[ML_x]^{y+}$ □ $M^{y+} + xL$ and equilibrium constant of this dissociation equilibrium is known as Instability constant which is a measure of stability. The stability of complex also depends upon EAN of central metal atom, charge on central metal atom, basic nature of ligand, chelation, and nature of metal ion and ligand.

Q12. The effective atomic number of Co in $Co(CO)_4$ is 35 and hence it is less stable. It attains stability by -

(A) oxidation of Co

(B) reduction of Co

(C) dimerization

(D) tetra merization

Q13. Which one of the following does not follow EAN rule -

(A) $Fe(CO)_5$

(B) $V(CO)_6$

(C) $K_4[Fe(CN)_6]$

(D) $Mn_2(CO)_{10}$

Q.14 Which complex is most stable -

(A) $[Cu(CN)_2]^-$ $K_d = 1 \times 10^{-16}$

(B) $[Fe(CN)_6]^{4-}$ $K_d = 1 \times 10^{-37}$

(C) $[Fe(CN)_6]^{3-}$ $K_d = 1 \times 10^{-44}$

(D) $[Ag(CN)_2]^-$ $K_d = 1 \times 10^{-20}$

Paragraph for Question No. 15 to 19

Valence bond theory successfully explains the magnetic behaviour of complexes. The substances which contains unpaired electrons are paramagnetic and paramagnetic character increases as the number of unpaired electrons increases, Magnetic moment of a complex can be determined experimentally and by using formula $\sqrt{n(n+2)}$ and we can detemine the number of unpaired electrons in it. This information is important in writing the electronic structure of complex which in turn also useful in deciding the geometry of complex.

Q15. The magnetic moments (spin only) of $[Ni(Cl)_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ are -

(A) 0, 0

(B) 2.82 B. M., 2.82 B. M.

(C) 0, 2.82 B. M.

(D) 2.82 B. M., 0

Q16. There are four complexes of Ni. Select the complex/es which will be attracted by magnetic field

$[Ni(CO)_4]^{2-}$	$[NiCl_6]^{4-}$	$Ni(CO)_4$	$[Ni(H_2O)_6]^{2+}$
I	II	III	IV

(A) I only (B) II and IV only

(C) II, III and IV (D) II and IV

Q17. The magnetic moments of complexes given below are in the order.

$Ni(CN)_4^{2-}$,	$[Mn(CN)_4]^{2-}$,	$[Cr(NH_3)_6]^{3+}$	$[CoF_6]^{3-}$
I	II	III	IV

(A) I > II > III > IV (B) I < II < III < IV

(C) IV > II > I > III (D) IV < II < I < III

Q18. The magnetic moment of $[Cu(NH_3)_4]^{2+}$ was found to be 1.73 B. M. The number of unpaired electrons in the complex is -

(A) 0 (B) 1

(C) 2 (D) 3

Q19. Which of the following complexes is a paramagnetic -

(A) $[Fe(CN)_6]^{4-}$

(B) $[Ni(H_2O)_6](NO_3)_2$

(C) $[Co(NH_3)_6]Cl_3$

(D) $[Pt(NH_3)_4]Cl_2$

Paragraph for Question No. 20 to 22

The chief factor responsible for the stability of the chelate ring is the entropy change which can be viewed by probability factor. As per probability factor when monodentate ligands are substituted by the bidentate ligands entropy changes is positive. Stability constants of the complexes are indicated by the terms K_{stab}

For the given complexes; K_{stab} values are given as follows.

Complex	**logK_{stab}**
$Fe(NH_3)_6^{2+}$	4
$Co(NH_3)_6^{2+}$	6
$Fe(en)_3^{2+}$	8

en is ethylene diamine

Q20. What is the log K_{stab} of $Co(en)_3^{2+}$ -

(A) 14 (B) 8

(C) 7 (D) 5

Q21. Regarding substitution reaction of $Fe(NH_3)_6^{2+}$ by en which of the following is/are correct ?

(A) In the substitution reaction $\Delta_r S°$ is positive

(B) In the subrtitution reaction $\Delta_r S°$ is negative

(C) Co-N bond strength in both the complexes are almost

(D) Both (A) & (C) are correct

Q22. For the substitution reaction of $[Co(NH_3)_6]^{3+}$ by en determine the standard free change ($\Delta_r G°$) at 300 K-

(A) –46 KJ/mol

(B) –70.4 KJ/mol

(C) –90 KJ/mol

(D) –28.42 KJ/mol

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q23. Match List I with List II and Select the correct answer using the codes given below the lists:

List 1	**List 2**
Complex Ions	Magnetic Moment µm Bohr Magnetons
a. $[Fe(CN)_6]^{-4}$	1. 1.73
b. $[Ti(H_2O)_6]^{3+}$	2. 5.93
c. $[Cr(H_2O)_6]^{3+}$	3. 0.00
d. $Ni(H_2O)_6]^{2+}$	4. 2.83
e. $[FeF_6]^{-3}$	5. 3.88

Q24. Match the complex ion given in column I with the hybridisation and Number of unpaired e^- given in column II and assign the correct code :

	Column I	Column II Hybridisation, Unpairide	
(a)	$[Cr(H_2O)_6]^{3+}$	p.	dsp^2, 1
(b)	$[Co(CN)_4]^{2-}$	q.	sp^3d^2, 5
(c)	$[Ni(NH_3)_6]^{2+}$	r.	d^2sp^3, 3
(d)	$[Mn\ F_6]^{4-}$	s.	sp^3, 4
		t.	sp^3d^2, 2

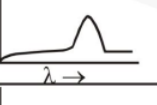
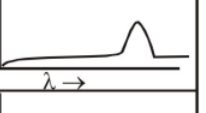
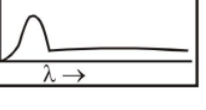
Q25.

Column-I	Column-II
(A) $[Ni(CN)_4]^{2-}$	(P) Outer orbital complex
(B) $[Co(NH_3)_6]^{2+}$	(Q) Low spin complex
(C) $[Fe(H_2O)_6]^{3+}$	(R) Inner orbital complex
(D) $Ni(CO)_4$	(S) Paramagnetic

Q26. Match the column :

Column-I	Column-II
(A) $[CuCl_4]^{2-}$	(P) Trigonal bi pyramidal
(B) $[Cu(NH_3)_4)^{2+}$	(Q) Octahedral
(C) $[Cr(NH_3)_6]^{3+}$	(R) Square planer
(D) $[Fe(CO)_5]$	(S) Tetrahedral

Q27. Match the following :

(A)	$[Co\ (NH_3)6]^{3+}$	(P)	
(B)	$[Co\ (NH_3)_4en]^{3+}$	(Q)	
(C)	$[Co\ (en)_3]^{3+}$	(R)	
(D)	$[Co\ \ Cl_6]^{3-}$	(S)	

TRIPLE MATCH TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Q28.

Column-I	Column-II	Column-III
(A) $[NICl_4]^{-2}$	(P) $\mu = 0$	(w) Inner orbital complex
(B) $[Ni(CO)_4]$	(Q) $\mu = \sqrt{8}$	(x) Outer orbital complex
(C) $[Ni(CN)_4]^{-2}$	(R) Paramagnetic	(y) sp^3
(D) $[NiF_6]^{-2}$	(S) Diamagnetic	(z) dsp^2

The correct match set of the given matrix.

(A) A, R, z (B) D, R, w (C) B, P, y (D) C, S, w

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q.29 How many structural isomers are possible for the formula $CrCl_3 \cdot 6H_2O$

Q30. How many hydrated isomers are possible for the formula $CrCl_3 \cdot 3NH_3 \cdot H_2O$ in octahedral complex.

Q31. Identify the complexes. How many are expected to be coloured ?

(i) $[Ti(NO_3)_4]$

(ii) $[Cu(NCCH_3)_4]^+ BF^-$

(iii) $[Cr(NH_3)_6]^{3+} 3Cl^-$

(iv) $K_3[VF_6]$

ASSERTION AND REASON

These questions contains, Statement-I (assertion) and Statement-II (reason).

(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I.

(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I.

(C) Statement-I is True, Statement-II is False.

(D) Statement-I is False, Statement-II is True.

Q32. **Statement-I :** Hydrazine is a neutral ligand.
Statement-II : It has two N as donor atoms and behaves as a chelating ligand.

Q33. **Statement-I :** Cis - $[Fe(en)_2Cl_2]^+$ can form recemic mixture.
Statement-II : Cis - $[Fe(en)_2Cl_2]^+$ is optically active square planar complex .

Q34. **Statement-I :** $[Ti(H_2O)_6]^{4+}$ is coloured while $[Sc(H_2O)_6]^{3+}$ is colourless.
Statement-II : d-d transition is not possible in $[Sc(H_2O)_6]^{3+}$.

Q35. **Statement-I :** $K_2[PtCl_6]$ gives white ppt when reacts with $AgNO_3$.
Statement-II : Chloride ion in the complex is non-ionisable.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
D	C	A	A	D	B	B	C	C	C
11	12	13	14	15	16	17	18	19	20
D	B	A	C	B	B	C	B	D	C
21	22	23	24	25	26	27	28	29	30
C	C	B	D	B	A	B	C	B	B
31	32	33	34	35	36	37	38	39	40
B	A	C	D	D	C	D	C	C	B
41	42	43	44	45	46	47	48	49	50
B	D	A	B	B	D	D	C	C	A
51	52	53	54	55	56	57	58	59	60
A	B	A	B	A	D	A	C	B	B
61	62	63							
D	A	A							

Ex-1A

1. B,C,D **2.** B,C,D **3.** A,B,C **4.** B,C,D **5.** A,B,D

6. B,C,D **7.** A,D **8.** B,C,D **9.** A,B,C **10.** B, D

11. A,B **12.** B **13.** C **14.** D **15.** A

16. C **17.** C

18. (A) - i, iv (B)-ii (C)-ii, iii (D)-i, iv

19. (A) - iv (B)- i, ii, iv (C)-iii, iv (D)-i, ii, iv

20. (i) $\rightarrow$ s, (ii) $\rightarrow$ p, (iii) $\rightarrow$ q, (iv) $\rightarrow$ r,t

21. 9 **22.** 6 **23.** 1 **24.** 4 **25.** 3

26. 15 **27.** 0 **28.** 3 **29.** 8 **30.** B

31. A **32.** A **33.** C **34.** C

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	B	D	D	A	B	C	A	B	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	B	A	C	B	B	D	D	D	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	D	D	C	B	B	B	D	A	A
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
A	D	D	A	D	A	A	A	A	D
41	**42**	**43**	**44**	**45**	**46**				
C	D	C	C	C	A				

Ex-2A

1. A,C,D **2.** B,C **3.** A,C,D **4.** B,C,D **5.** A,B

6. C,D **7.** A,C,D **8.** A,B,C **9.** A,B **10.** A,B

11. A,B,D **12.** C **13.** B **14.** C **15.** D

16. D **17.** B **18.** B **19.** B **20.** A

21. A **22.** A **23.** B **24.** A $\rightarrow$ R, B $\rightarrow$ P, C $\rightarrow$ T, D $\rightarrow$ Q

25. A $\rightarrow$ Q,R B $\rightarrow$ Q,R, S, C $\rightarrow$ P,S, D $\rightarrow$ P,Q **26.** A $\rightarrow$ R, B $\rightarrow$ S, C $\rightarrow$ Q, D $\rightarrow$ P

27. A $\rightarrow$ Q, B $\rightarrow$ P, C $\rightarrow$ S, D $\rightarrow$ R **28.** C **29.** 3 **30.** 0

31. 2 **32.** C **33.** C **34.** D **35.** D